

Class 08 Mini-Project: Unsupervised Learning Analysis of Human Breast Cancer Cells

Seunghoon Oh (PID: A19372132)

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Exploratory data analysis

Data preparation

```
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)
```

```
head(wisc.df, 4)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001		0.14710
842517	0.08474	0.07864	0.0869		0.07017
84300903	0.10960	0.15990	0.1974		0.12790
84348301	0.14250	0.28390	0.2414		0.10520
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419		0.07871	1.0950	0.9053
842517	0.1812		0.05667	0.5435	0.7339
84300903	0.2069		0.05999	0.7456	0.7869
84348301	0.2597		0.09744	0.4956	1.1560
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003		0.006193	25.38	17.33
842517	0.01389		0.003532	24.99	23.41
84300903	0.02250		0.004571	23.57	25.53
84348301	0.05963		0.009208	14.91	26.50
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0	0.1622		0.6656
842517	158.80	1956.0	0.1238		0.1866
84300903	152.50	1709.0	0.1444		0.4245
84348301	98.87	567.7	0.2098		0.8663
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
84348301	0.6869	0.2575	0.6638		
	fractal_dimension_worst				
842302	0.11890				
842517	0.08902				
84300903	0.08758				
84348301	0.17300				

```
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
# Create diagnosis vector for later
diagnosis <- wisc.df$diagnosis
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
dim (wisc.data)
```

```
[1] 569  30
```

```
nrow (wisc.data)
```

```
[1] 569
```

=> total of 569 observations.

Q2. How many of the observations have a malignant diagnosis?

```
nrow(wisc.df[wisc.df$diagnosis=="M", ])
```

```
[1] 212
```

=> 212 observations have a malignant diagnosis.

Or you can use `table()` function to do this.

```
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
length(grep("_mean", colnames(wisc.data), value=FALSE))
```

```
[1] 10
```

Principal Component Analysis

Performing PCA

```
# Check column means and standard deviations
colMeans(wisc.data)
```

radius_mean	texture_mean	perimeter_mean
1.412729e+01	1.928965e+01	9.196903e+01
area_mean	smoothness_mean	compactness_mean
6.548891e+02	9.636028e-02	1.043410e-01
concavity_mean	concave.points_mean	symmetry_mean
8.879932e-02	4.891915e-02	1.811619e-01
fractal_dimension_mean	radius_se	texture_se
6.279761e-02	4.051721e-01	1.216853e+00
perimeter_se	area_se	smoothness_se
2.866059e+00	4.033708e+01	7.040979e-03
compactness_se	concavity_se	concave.points_se
2.547814e-02	3.189372e-02	1.179614e-02
symmetry_se	fractal_dimension_se	radius_worst
2.054230e-02	3.794904e-03	1.626919e+01
texture_worst	perimeter_worst	area_worst
2.567722e+01	1.072612e+02	8.805831e+02
smoothness_worst	compactness_worst	concavity_worst
1.323686e-01	2.542650e-01	2.721885e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
1.146062e-01	2.900756e-01	8.394582e-02

```
apply(wisc.data,2,sd)
```

radius_mean	texture_mean	perimeter_mean
3.524049e+00	4.301036e+00	2.429898e+01
area_mean	smoothness_mean	compactness_mean
3.519141e+02	1.406413e-02	5.281276e-02

concavity_mean	concave.points_mean	symmetry_mean
7.971981e-02	3.880284e-02	2.741428e-02
fractal_dimension_mean	radius_se	texture_se
7.060363e-03	2.773127e-01	5.516484e-01
perimeter_se	area_se	smoothness_se
2.021855e+00	4.549101e+01	3.002518e-03
compactness_se	concavity_se	concave.points_se
1.790818e-02	3.018606e-02	6.170285e-03
symmetry_se	fractal_dimension_se	radius_worst
8.266372e-03	2.646071e-03	4.833242e+00
texture_worst	perimeter_worst	area_worst
6.146258e+00	3.360254e+01	5.693570e+02
smoothness_worst	compactness_worst	concavity_worst
2.283243e-02	1.573365e-01	2.086243e-01
concave.points_worst	symmetry_worst	fractal_dimension_worst
6.573234e-02	6.186747e-02	1.806127e-02

```
# Perform PCA on wisc.data by completing the following code
wisc.pr <- prcomp(wisc.data, scale = TRUE)
# Look at summary of results
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					

Cumulative Proportion	1.00000	1.00000
-----------------------	---------	---------

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

=> 44.27% is the proportion of the original variance captured by PC1.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

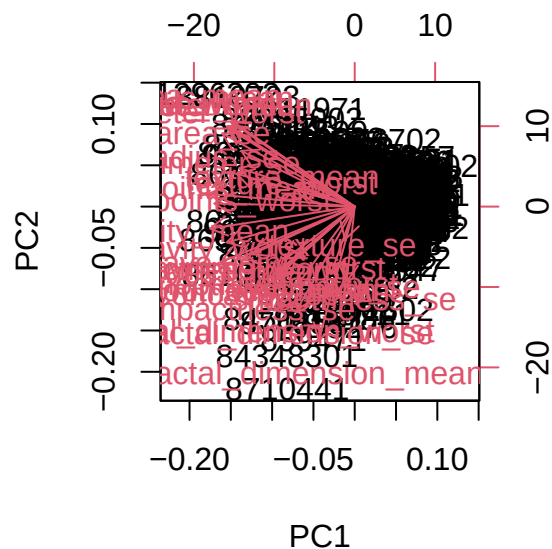
=> Three PCs (PC1-PC3).

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

=> Seven PCs (PC1-PC7).

Interpreting PCA results

```
biplot(wisc.pr)
```

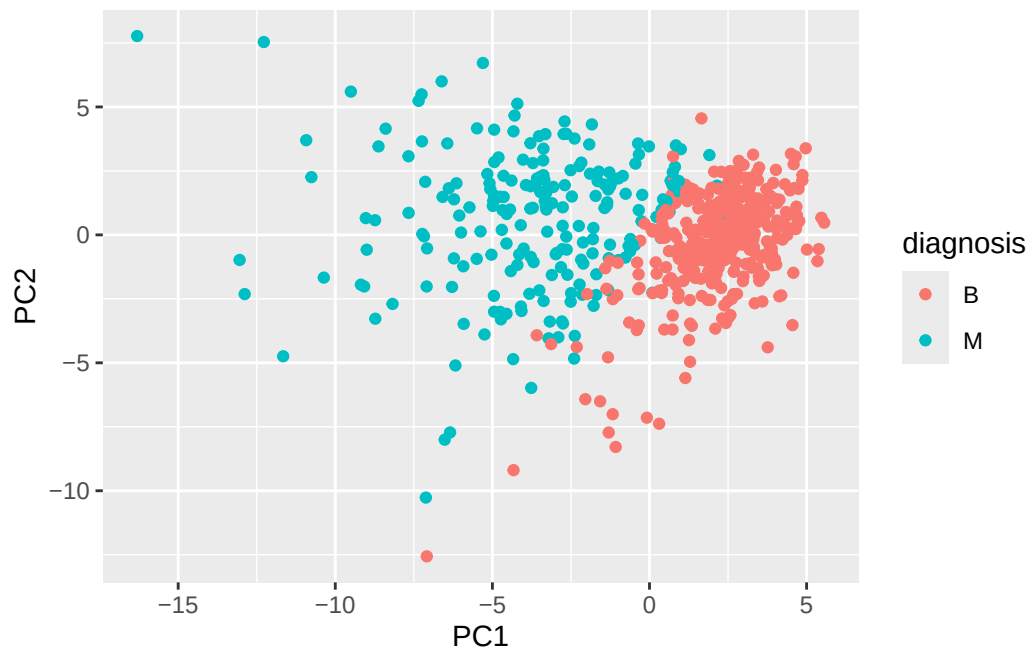


Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

Difficult to understand. All the observations (row names) and column names are shown in the plot as text. The data is so cluttered in this plot.. which makes it difficult to interpret.

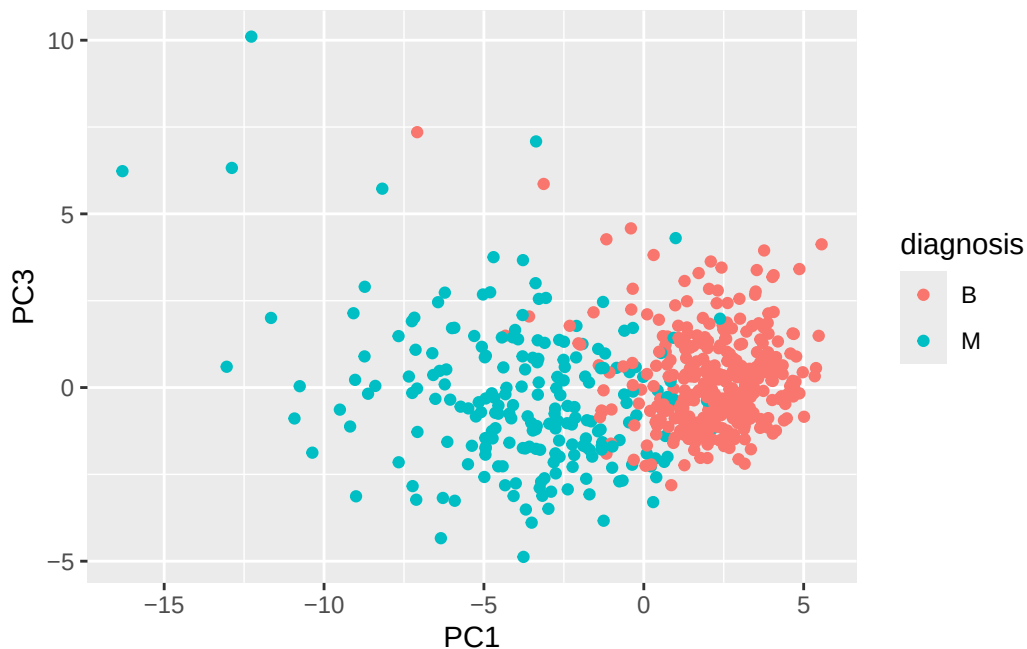
```
# Scatter plot observations by components 1 and 2
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



Compared to PC1-PC2 ggplot, the PC1-PC3 ggplot shows more overlap between benign and malignant datapoint area. PC1 shows more separation between datapoints from different groups than PC2 and PC3 - which aligns with the PC1-3 definition.

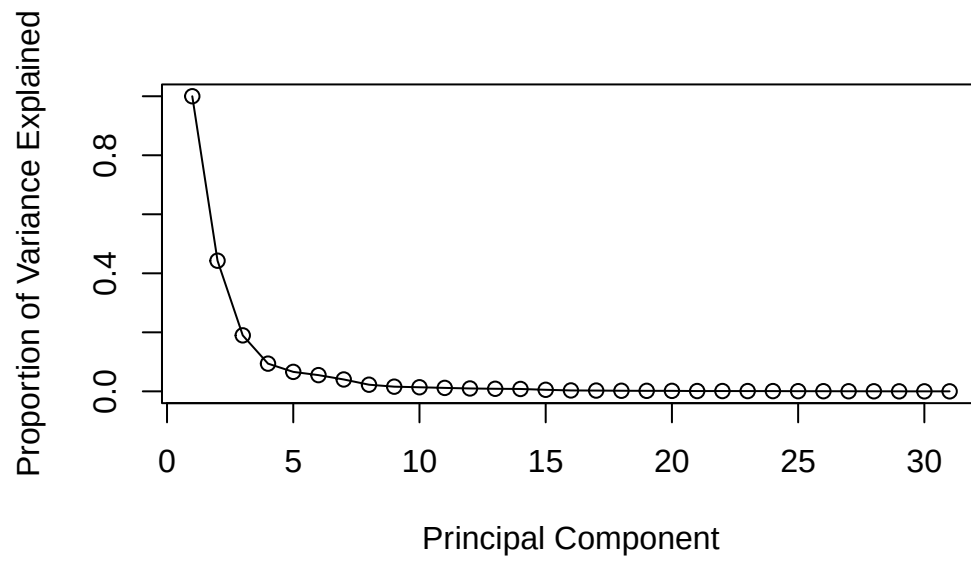
Variance explained

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

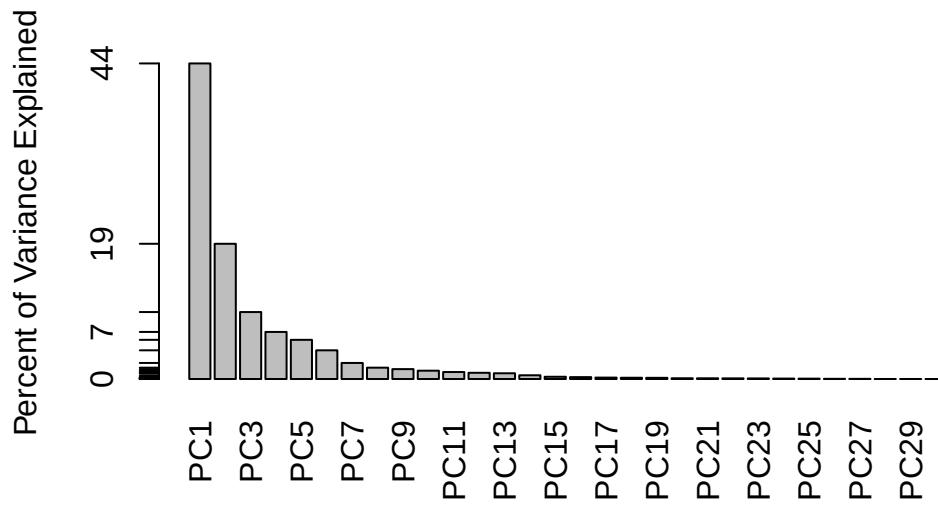
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```

```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



```
## ggplot based graph
##install.packages("factoextra")
##library(factoextra)
##fviz_eig(wisc.pr, addlabels = TRUE)
```

Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

The loading value for PC1 is approximately -0.26. Considering the sign, this feature being high is associated with low PC1 scores.

```
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se
0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

From the result of this code, we are able to see that `concave.points_mean` has the highest magnitude of loading value -> which means that this component is most closely associated to the PC1.

Hierarchical clustering

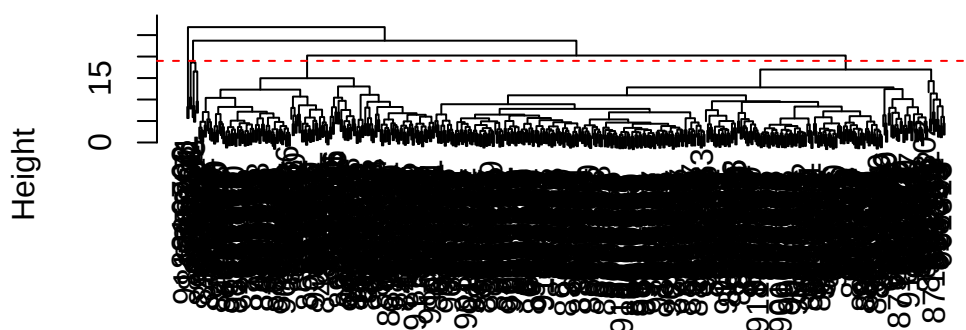
```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
```

```
wisc.hclust <- hclust(data.dist, method="complete")
```

Results of hierarchical clustering

```
plot(wisc.hclust)
abline(h=19, col="red", lty=2)
```

Cluster Dendrogram



```
data.dist
hclust (*, "complete")
```

Selecting number of clusters

```
wisc.hclust.clusters <- cutree (wisc.hclust, k=4)
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Q11. OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

```

for (k in 2:6) {
  cat("\nk =", k, "\n")
  print(table(cutree(wisc.hclust, k = k), diagnosis))
}

```

```

k = 2
  diagnosis
    B    M
1 357 210
2   0   2

```

```

k = 3
  diagnosis
    B    M
1 355 205
2   2   5
3   0   2

```

```

k = 4
  diagnosis
    B    M
1  12 165
2   2   5
3 343  40
4   0   2

```

```

k = 5
  diagnosis
    B    M
1  12 165
2   0   5
3 343  40
4   2   0
5   0   2

```

```

k = 6
  diagnosis
    B    M
1  12 165
2   0   5
3 331  39

```

4	2	0
5	12	1
6	0	2

=> a good clustering can be defined as a clustering such that each clusters are containing mostly M or mostly B diagnosed observations. When we change k from 2 to 6, a meaningful change (threshold) is shown between k=3 and k=4, since from k=4 clusters are divided into “most M” or “most B” clusters. Thus, k=4 is a good clustering among these options.

Using different methods

Q12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

single

```
wisc.hclust_s <- hclust(data.dist, method="single")
for (k in 2:6) {
  cat("\nk =", k, "\n")
  print(table(cutree(wisc.hclust_s, k = k), diagnosis))
}
```

```
k = 2
diagnosis
  B  M
1 357 210
2   0   2
```

```
k = 3
diagnosis
  B  M
1 356 210
2   1   0
3   0   2
```

```
k = 4
diagnosis
  B  M
1 356 209
```

2	1	0
3	0	2
4	0	1

```
k = 5
  diagnosis
    B    M
1 356 209
2   1   0
3   0   1
4   0   1
5   0   1
```

```
k = 6
  diagnosis
    B    M
1 356 208
2   0   1
3   1   0
4   0   1
5   0   1
6   0   1
```

average

```
wisc.hclust_a <- hclust(data.dist, method="average")
for (k in 2:6) {
  cat("\nk =", k, "\n")
  print(table(cutree(wisc.hclust_a, k = k), diagnosis))
}
```

```
k = 2
  diagnosis
    B    M
1 357 209
2   0   3
```

```
k = 3
  diagnosis
    B    M
1 355 209
```

2	2	0
3	0	3

```
k = 4
  diagnosis
    B    M
1 355 209
2   2   0
3   0   1
4   0   2
```

```
k = 5
  diagnosis
    B    M
1 355 208
2   2   0
3   0   1
4   0   2
5   0   1
```

```
k = 6
  diagnosis
    B    M
1 355 202
2   0   6
3   2   0
4   0   1
5   0   2
6   0   1
```

ward.D2

```
wisc.hclust_w <- hclust(data.dist, method="ward.D2")
for (k in 2:6) {
  cat("\nk =", k, "\n")
  print(table(cutree(wisc.hclust_w, k = k), diagnosis))
}
```

```
k = 2
  diagnosis
    B    M
```



```

1  20 164
2 337  48

k = 3
  diagnosis
    B    M
1    0 115
2   20  49
3 337  48

k = 4
  diagnosis
    B    M
1    0 115
2    6  48
3 337  48
4   14   1

k = 5
  diagnosis
    B    M
1    0  59
2    0  56
3    6  48
4 337  48
5   14   1

k = 6
  diagnosis
    B    M
1    0  59
2    0  56
3    6  48
4 235  46
5 102   2
6   14   1

```

Based on these results, **single** or **average** method doesn't give meaningful clustering in **k** in 2:6 range, while **ward.D2** gives meaningful clustering when **k=6**.

Comparing **complete** and **ward.D2**:

```
cat("complete\n")
```

complete

```
print(table(cutree(wisc.hclust, k = 6), diagnosis))
```

	diagnosis	
	B	M
1	12	165
2	0	5
3	331	39
4	2	0
5	12	1
6	0	2

```
cat("\nward.D2\n")
```

ward.D2

```
print(table(cutree(wisc.hclust_w, k = 6), diagnosis))
```

	diagnosis	
	B	M
1	0	59
2	0	56
3	6	48
4	235	46
5	102	2
6	14	1

This result seems to be pointing that ward.D2 is slightly better in k=6.

Combining methods

Clustering on PCA results

```
# Use the first 7 PCs, which explain at least 90% of the variance
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
```

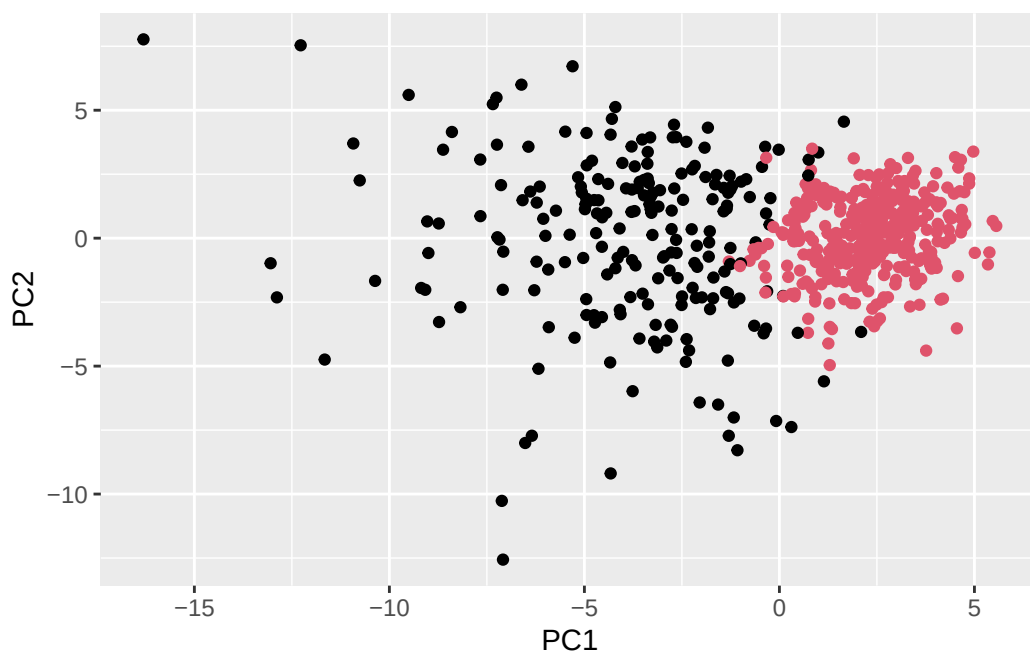
```
grps <- cutree(wisc.pr.hclust, k = 2)
table(grps)
```

```
grps
  1  2
216 353
```

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1  28 188
  2 329  24
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k=2)
```

Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

```
# Compare to actual diagnoses
table(wisc.pr.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.pr.hclust.clusters	B	M
1	28	188
2	329	24

It does a quite good job in clustering, to separate out the two diagnoses - with only two groups.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

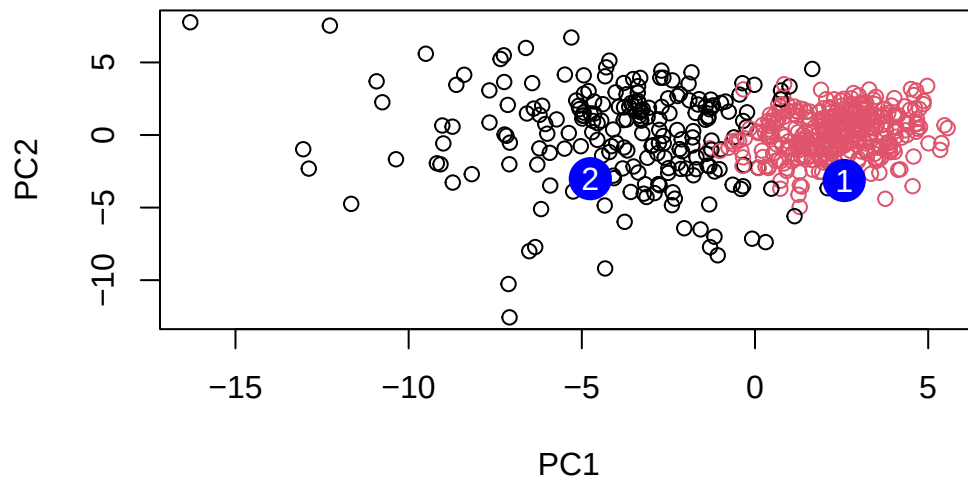
Just compared from the summary results generated by `table()`, using PCA for clustering seems to be yielding better results than hierarchical clustering with less number of groups (`k`).

Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Group 2, as the same PCA results on our original dataset showed that left cluster in this PC1-PC2 graph has “malignant” diagnosis.